

Tutorial Week 3

- Properties of pure substances

Q1) Find the saturation pressure for water if its temperature is measured to be 100°C . Also, find the specific volume, enthalpy, and internal energy for the saturated liquid at this temperature.

Q2) Find the saturation temperature for R-134a if its pressure is measured to be 500 kPa. Also, find the specific volume, enthalpy, and internal energy of the saturated vapour at this pressure.

Q3) The temperature in a vessel filled with water is measured to be 222.5°C . Using the interpolation method, find the saturation pressure at this temperature.

Q4) A container is filled with superheated steam at a pressure of 0.3 MPa and a temperature of 325°C . Determine the steam's specific volume, specific internal energy, and specific enthalpy.

Q5) The temperature in a container filled with saturated water is measured to be 90°C . Determine its quality if the water has a specific volume of $2\text{ m}^3/\text{kg}$.

Q6) The temperature of water in a piston–cylinder is measured to be 200°C , and its pressure is measured to be 400 kPa. Determine which phase the water is in, and which table should be used to find its properties. What table should be used if the pressure is maintained at 400 kPa, but the temperature is lowered to 100°C ?

Q7) 10 kg of water fills a 1.115 m^3 rigid container at an initial temperature of 150°C . The container is then heated until the pressure is 2000 kPa. Determine the initial pressure and quality, and the final temperature.

Q8) A piston-cylinder device contains 0.6 kg of steam at 300°C and 0.5 MPa. The steam is cooled at a constant pressure until one-half of the mass condenses. Show the process on a T-v diagram, find the final temperature, and determine the volume change.

- Ideal Gas

Where not otherwise stated, assume that atmospheric pressure is 101.325 kPa, and give all relevant answers in terms of absolute pressure.

Q9) The pressure gauge on a 3-m^3 oxygen tank reads 600 kPa. Determine the amount of oxygen in the tank if the temperature is 28°C and the atmospheric pressure is 97 kPa.

Q10) An automobile tire with a volume of 0.5 m^3 is inflated to a gauge pressure of 250 kPa. Calculate the mass of air in the tire if the temperature is 25°C using the ideal-gas equation of state.

Q11) A spherical balloon with a diameter of 10 m is filled with helium at 25°C and a gauge pressure of 250 kPa. Determine the mole number and the mass of the helium in the balloon.

Q12) A rigid tank with an unknown volume is divided into two parts by a partition. One side of the tank contains an ideal gas at 900°C. The other side is evacuated and has a volume three times the size of the part containing the gas. The partition is now removed, and the gas expands to fill the entire tank. Heat is now transferred to the gas until the pressure equals the initial pressure. Determine the final temperature of the gas.

Q13) A 2-m³ tank containing air at 10°C and a gauge pressure of 300 kPa is connected through a valve to another tank containing 3 kg of air at 35°C and 100 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 25°C. Determine the volume of the second tank and the final equilibrium pressure of air. Give your answer in terms of absolute pressure.

Q14) A piston-cylinder arrangement contains 0.04 m³ of air at 50°C and at an absolute pressure of 500 kPa. Heat is added in the amount of 60 kJ and work is done by a paddle wheel until the temperature reaches 600°C. If the pressure is held constant, how much paddle-wheel work must be added to the air? Assume constant specific heats.